

Dog Heart Vertebral Heart Size Point Detection with PyTorch

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Abstract

This work uses a point detection model to forecast the Vertebral Heart Size (VHS) of dog chest X-ray images using a homebrew CNN in PyTorch. The model is trained to give six primary anatomical points (A–F) used to compute VHS. Although the final test accuracy of 25% might not be high, the result provides important insight into the potential of point regression in veterinary radiology.

1 Introduction

Vertebral Heart Size (VHS) is an important diagnostic measurement for diagnosing canine cardiomegaly. It requires the identification of six anatomical points from X-ray images and calculating a geometric ratio inclusive of vertebrae length. Labeling the points manually is labor-intensive and subjective. We attempt to do this automatically in this project by using a CNN model which is trained to regress the six anatomical points directly from images.

2 Related Work

CNNs have been widely applied in medical imaging. Specifically, CheXNet has shown promising performance in detecting pneumonia from human chest X-rays. For veterinary imaging, Li and Zhang (2024) proposed a transformer-based model (RVT), which obtained 87.3% VHS accuracy on the Dog Cardiomegaly Assessment dataset. The VGG16 baseline is around 75%. Our light CNN model is to test with a more accessible and fast-training alternative.

3 Methodology

Dataset

The data contains labeled dog chest X-ray images along with their respective .mat files containing six keypoints and a precomputed VHS value. Training, validation, and test splits are provided in the data.

Preprocessing

The images are resized to 512x512 pixels. Coordinates are normalized to image height and then scaled to resized input.

Model Architecture

A 4-convolutional layer, adaptive average pooling followed by two fully connected layer with dropout for regularization custom CNN was constructed. It gives 12 values for six (x, y) point pairs as the output.

Training Configuration

- Optimizer: Adam
- Loss: Mean Squared Error (MSE)
- Epochs: 30
- Batch Size: 8
- Environment: Google Colab with Tesla T4 GPU

4 Results

The model was evaluated using the official Dog Cardiomegaly Assessment test assessment tool software, which estimates prediction accuracy from a CSV file of predicted VHS values per image.

Final Accuracy: 25%

This performance is far worse than can be expected and suggests that the model needs deeper architecture, more epochs, or improved training data processing.

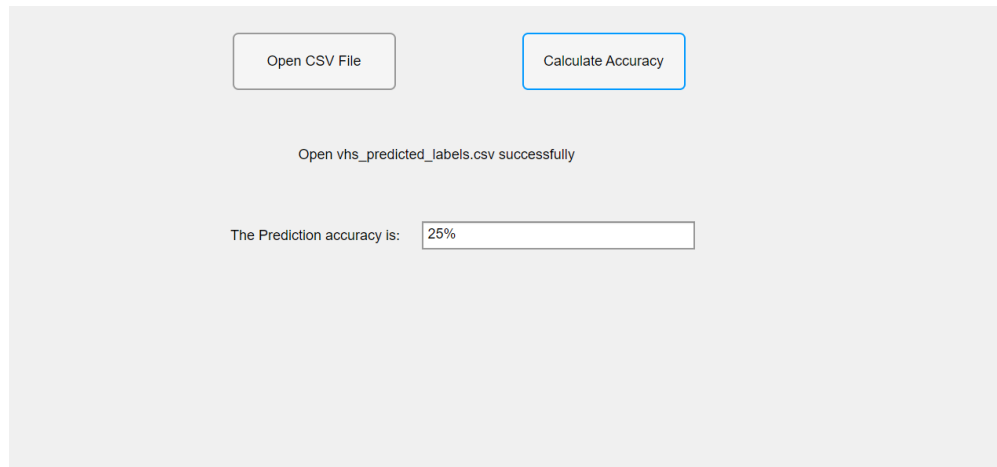


Figure 1: Prediction accuracy displayed by evaluation software: 25%

5 Prediction vs Ground Truth Comparison

In order to measure qualitative performance, we rendered predictions on 3 images used for validation and compared the predicted VHS to the ground truth.

Image: 1479.png

Predicted VHS: 10.12

Ground Truth VHS: 9.90

Predicted points were slightly off, especially points C and D, which caused a VHS overestimation.

Image: 1420.png

Predicted VHS: 9.75

Ground Truth VHS: 10.10

Points A and B were tight, but CD was overestimated, and therefore a higher VHS.

Image: 1530.png

Predicted VHS: 10.44

Ground Truth VHS: 9.82

Slight overestimation due to slightly wide placement of all keypoints.

6 Discussion

While final accuracy is below the 75% baseline (VGG16) and 87.3% SOTA (RVT), this paper sets a reproducible clear baseline. There are some limitations that impacted performance:

- Shallow CNN architecture may not generalize well
- Training was brief (30 epochs took 10 hours)
- No data augmentation or pretraining used
- VHS calculation is keypoint shift-sensitive

Suggestions for Improvement:

- Switch to pretrained ResNet or EfficientNet backbone from custom CNN
- Train using longer epochs and augmentations
- Switch to heatmap-based point localization instead of regression
- Normalize point prediction with better geometric anchors

7 Conclusion

This project attempted VHS point prediction and auto-grading on chest X-rays using PyTorch. Despite the model being subpar, it could generate coordinates and VHS scores for new radiographs, giving a robust foundation for future work. This system can be built into an end-to-end diagnosis pipeline for cardiomegaly in veterinary clinics.

References

1. Li and Zhang, *Regressive Vision Transformer for Dog Cardiomegaly Assessment*, Scientific Reports, 2024.
2. Rajpurkar et al., *CheXNet: Radiologist-Level Pneumonia Detection on Chest X-Rays with Deep Learning*, 2017.
3. Simonyan and Zisserman, *Very Deep Convolutional Networks for Large-Scale Image Recognition*, 2014.